



DJANGO CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & PROJECT

Installation

```
pip install django
django-admin startproject mysite
python manage.py startapp blog
python manage.py runserver
python manage.py migrate
```

INSTALLED_APPS must include new apps

04. URLs & ROUTING

Workflow

```
# urls.py
urlpatterns = [
    path('posts/', views.PostListView.as_view()),
    path('posts/<int:pk>/', views.PostDetailView...
    path('api/', include('api.urls')),
]
re_path(r'^old/(?P<slug>[-w]+)/$', view)
```

02. MODELS

Architecture

```
from django.db import models
class Post(models.Model):
    title = models.CharField(max_length=200)
    body = models.TextField()
    created = models.DateTimeField(auto_now_add=True)
    author = models.ForeignKey(User, on_delete=mo...
    class Meta: ordering = ['-created']
```

05. ORM QUERYSETS

CRUD API

```
Post.objects.all()
Post.objects.filter(author=user, published=Tr...
Post.objects.exclude(status='draft')
Post.objects.get(pk=1) # raises DoesNotExist
Post.objects.order_by('-created')[:10]
Post.objects.select_related('author')
Post.objects.prefetch_related('tags')
```

03. VIEWS

YAML / Config

```
from django.views import View
class PostListView(View):
    def get(self, request):
        posts = Post.objects.all()
        return render(request, 'list.html', {'posts': po...
    # Function-based:
    def post_list(request):
        return JsonResponse({'posts': list(...)}, saf...
```

06. MIGRATIONS

Integration

```
python manage.py makemigrations
python manage.py migrate
python manage.py showmigrations
python manage.py migrate blog 0002
python manage.py sqlmigrate blog 0001
```

Never delete migrations in production



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. TEMPLATES

Hardening

```
{{ variable }} {# comment #}  
{% if user.is_authenticated %}...{% endif %}  
{% for post in posts %}...{% endfor %}  
{% extends 'base.html' %}  
{% block content %}...{% endblock %}  
{% url 'post-detail' pk=post.pk %}
```

10. MIDDLEWARE

Unit Tests

```
MIDDLEWARE = [  
    'django.middleware.security.SecurityMiddlewar...  
    'django.contrib.sessions.middleware.SessionMi...  
    'django.middleware.csrf.CsrfViewMiddleware',  
    'myapp.middleware.CustomMiddleware',  
]  
Custom: define __init__(get_response) and __call__(request)
```

08. ADMIN

Observability

```
from django.contrib import admin  
@admin.register(Post)  
class PostAdmin(admin.ModelAdmin):  
    list_display = ['title', 'author', 'created']  
    search_fields = ['title', 'body']  
    list_filter = ['status', 'created']  
python manage.py createsuperuser
```

11. FORMS & VALIDATION

Commands

```
class PostForm(forms.ModelForm):  
    class Meta:  
        model = Post  
        fields = ['title', 'body']  
    form = PostForm(request.POST or None)  
    if form.is_valid(): form.save()  
    def clean_title(self): # custom validation
```

09. AUTHENTICATION

Optimization

```
from django.contrib.auth.decorators import login_required  
@login_required(login_url='/login/')  
def dashboard(request): ...  
LoginView / LogoutView / PasswordResetView  
request.user.is_authenticated  
authenticate(username=u, password=p)  
login(request, user) / logout(request)
```

12. COMMON GOTCHAS

Warning

N+1: always use select_related/prefetch_related
CSRF token required in all POST forms
DEBUG=True exposes full tracebacks disable in prod
SECRET_KEY must be secret use env vars
Migrations in prod: always backup before migrate